Akhilesh Kumar Bhagat

Embedded Systems · HCI · Robotics — Student Research Assistant, IIT Kanpur (National Quantum Mission)

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SUMMARY

Electronic Systems student at IIT Madras and Student Research Assistant at IIT Kanpur under India's National Quantum Mission. I build human-centred hardware — real-time ESP32 and STM32 firmware, sensor interfacing, low-latency wireless links and Linux drivers — for musical interfaces, assistive devices and robots. Co-inventor on three Indian patents (two granted) and co-author of six papers at CHI, UIST, NIME, Creativity & Cognition and INDIS, including a Best Paper Award.

SKILLS

Languages	Embedded C, C, Python, Bash, Verilog, JavaScript
Microcontrollers	ESP32, ESP8266, STM32, Arduino, Seeeduino XIAO, Raspberry Pi & Pico (RP2040), NVIDIA Jetson
Protocols	MQTT, ESP-NOW, SPI, I^2C, UART, CAN, OSC, MIDI
Systems	FreeRTOS, Linux device drivers, ROS 2, Linux, Windows, macOS
Sensing & vision	OpenCV, Intel RealSense (RGB-D), IMUs, capacitive and gas sensors, eye tracking
Tools	Git, STM32CubeIDE, Arduino IDE, LTspice, Fusion, Blender, TouchDesigner, Max/MSP, Ollama

EDUCATION

Indian Institute of Technology Madras — B.S. in Electronic Systems — Sep 2023 – Sep 2027

Kendriya Vidyalaya Sukna — CBSE, Class I–XII — 2010 – 2023

EXPERIENCE

Student Research Assistant — IIT Kanpur · National Quantum Mission — Jul 2026 – Present · Kanpur

Teaching Assistant, School Connect — CODE, IIT Madras (part-time) — Dec 2025 – Present · Remote

Project Assistant — Indian Institute of Technology Madras — Apr 2026 – Jun 2026 · Chennai

Research Intern — HIVE Lab, IIT Kanpur — Nov 2024 – Mar 2026 · Kanpur

- Developed embedded systems and firmware for robotics, IoT and HCI prototypes on ESP32 and STM32; built and tested 10+ prototypes with sensor interfacing, actuator control and wireless communication.
- Co-inventor on 3 Indian patents (2 granted) and co-author of 6 papers at CHI, UIST, NIME, Creativity & Cognition and INDIS.
- Investigated large language model (LLM) integration for autonomous agents and decision-making systems.
- Ran microcontroller sessions as a TA and helped 20+ Master's and PhD students build their embedded prototypes.
- Worked with faculty and research scholars on experimental validation, system design and research dissemination.

Founder & Secretary — Yantra, Robotics Society of the IIT Madras BS programme — Aug 2024 – Jul 2026

- Founded and led the official robotics society: a 30-member core team and 1000+ members (Club Head 2024–25).
- Designed and delivered 30+ hands-on sessions on ESP32, STM32, Arduino and microcontroller programming.
- Organised workshops, hackathons and industry talks, and built the live control room for Robo Dominion, the society's robot battle.
- Mentored members on project execution, system design and research-oriented thinking.

Event Coordinator & Exhibitor — Techkriti 2025, IIT Kanpur — Mar 2025

- Demonstrated battery-powered, low-latency ESP32 wireless prototypes at a national festival of ~60,000 visitors.
- Coordinated public demos of NIME instruments — logistics, volunteers and live exhibits.

Group Lead — Saranda House, IIT Madras BS — Aug 2024 – Nov 2024

- Led collaborative learning and leadership activities for a cohort of 150+ students, raising peer participation by ~30%.

PATENTS

- Granted **Indian Patent No. 588076** — Wearable Musical Device for Simulating Indian Classical Percussion Audio (Tabla Glove · Appl. 202511071234)
- Granted **Indian Patent No. 587794** — A Musical Instrument (Swarings, ring-based MIDI controller · Appl. 202511072744)
- Filed **Appl. 202511070920** — Real-Time Visual Feedback System and Method for Minimizing the Risk of Jaywalking or Distracted Walking (PathBuddy · Nov 2025 · offered for licensing by IIT Kanpur)

PUBLICATIONS

- Best Paper **PathBuddy: A Smart Solution to Prevent Smartphone-Induced Jaywalking Risks with Enhanced Obstacle Awareness.** INDIS 2026, IIT Guwahati — first author.
- **When Assistive Technologies become Provocations: Unpacking Access in HCI practices using Crip Technoscience, Mouth Interfaces, and XR.** ACM CHI 2026.
- **Designing a Scaffolding for the Audience to Create Participatory Sense-making in Indian Classical Dance.** ACM Creativity & Cognition 2026.
- **Layika: A Wearable Device Mapping Traditional Hand Gestures to Tabla Sound.** NIME 2026.
- **Swarings: Exploring Expressive Freedom through Ring-Based Musical Interaction.** ACM UIST 2025 Adjunct.
- **GraviTone: A Tangible Musical Interface using Gravity Well for Sound and Music Creation.** NIME 2025.
- **India HCI 2025 posters & demos** — GraviTone; Swarings; The Tabla Glove: An Accompanist for Musical Training and Exploration.

PROJECTS

Research prototypes

Layika — Tabla Glove
ESP32 · FreeRTOS · ESP-NOW · MIDI
Conductive yarn on the finger creases detects taal taps; RTOS firmware sends hits over ESP-NOW to trigger tabla sounds live. Battery-powered. Patented; NIME 2026.

GraviTone
OpenCV · Max/MSP · TouchDesigner · OSC/MIDI
Balls orbit a gravity well on stretch fabric; real-time camera tracking of speed and position drives synths and visuals. Built the ball-tracking prototype. NIME 2025.

Mouth Tracker for XR
ESP32-CAM · IR camera · UART
Custom ESP32-CAM firmware and IR camera hardware (built on Project Babble) for hands-free XR control via tongue, cheek and jaw movement. CHI 2026.

HUX AI 2.0
Wearable cameras · computer vision · AI
Cap-mounted first-person camera plus a face camera, streamed together to study human experience. Designer, researcher and engineer.

Swarings
MIDI controller · touch & rotation sensing
Rings on a pipe: touch plays a note, sliding bends pitch, rotating modulates. Patented; UIST 2025; played live in NIME concerts.

PathBuddy
Depth sensing · Android · computer vision
External depth sensor warns phone users of obstacles with on-screen time-to-impact overlays. User study: more obstacles detected, lower cognitive load. Best Paper, INDIS 2026.

Reading Indian Classical Dance
Computer vision · multimodal AI · eye tracking
Recognises mudras and facial expressions in a performance and narrates their meaning; 10-viewer study with gaze tracking. Creativity & Cognition 2026.

Eye-Gaze Control for the ICU
ESP32 · Embedded C
Gaze-driven tile menu lets people in intensive care choose options and call for help; ESP32 drives the alerts, with blink filtering to avoid false triggers.

Embedded, robotics & AI

Robo Dominion Control Room
JavaScript · Web Serial · ESP-NOW · ESP32
Live broadcast page (camera feeds, scores, bot health) and an admin console that runs the match through an ESP-NOW master over Web Serial.

Linux Kernel Drivers
C · LKM · procfs/sysfs · Raspberry Pi
Loadable modules, a /proc driver with kernel-to-user data, and a GPIO LED driver on a Raspberry Pi Zero 2 W controlled from user space.

Facial Recognition Lock
Raspberry Pi · ESP32 · UART · OpenCV
Face recognition on the Pi; ESP32 firmware actuates the lock over a serial link.

Shape-Following Robot
Contour detection · JavaScript
Paths drawn on a touchscreen become drive commands the robot follows in real time.

Neuro — Multi-LLM Assistant
Python · LLMs · MQTT · ESP32
Routes plain-language requests to the right model or script: lookups, automation, image generation and IoT control.

Teleoperated Robots (Robo Rumble)
ESP8266/ESP32 · MQTT · WebSocket
Web-controlled robots over MQTT with bidirectional, low-latency control; run live with 150+ users.

Raspberry Pi Pico, Bare-Metal
C · Pico SDK · CMake
RP2040 programmed without an OS: GPIO, timing and low-level hardware interfacing.

Mini Sphere Bot
ESP32 · MQ gas sensors · motor control
Compact rolling robot that navigates and measures gas concentrations in hazardous spaces.

RGB-D 3D Object Detection
Python · Intel RealSense · OpenCV
Synchronised colour and depth capture places detections in real 3D space.

Digital Theremin
ESP8266 · capacitive sensing
Touch-free instrument that maps hand distance to pitch.